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Automatic HamNoSys Generation from Sign Language Images with Avatar-Based Verification

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MARCH 2026

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BONAFIDE CERTIFICATE

This is to certify that the project report entitled Automatic HamNoSys Generation from Sign Language Videos with Avatar-Based Verification submitted by S.Siva Pravallika(CB.EN.U4CSE22440), S.Indira Satya Sri (CB.EN.U4CSE22441), O.Sravani(CB.EN.U4CSE22457), and D.Vamsi Krishna(CB.EN.U4CSE22458) in partial fulfillment of the requirements for the award of Degree **Bachelor of Technology** in Computer Science and Engineering is a bonafide record of the work carried out under our guidance and supervision at the Department of Computer Science and Engineering, Amrita School of Computing, Coimbatore.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

We the undersigned solemnly declare that the project Automatic Generation and Verification of HamNoSys from Continuous Sign Language Videos Using Avatar-based Similarity Detection is based on our own work carried out during the course of our study under the supervision of Guide Ms. Nalinadevi K**,** Associate Professor, Computer Science and Engineering, and has not formed the basis for the award of any other degree or diploma, in this or any other Institution or University. In keeping with the ethical practice in reporting scientific information, due acknowledgement has been made wherever the findings of others have been cited.

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ABSTRACT

HamNoSys (Hamburg Notation System) is a popular notation system to write sign languages, which allows the language-free representation of both manual and non-manual features [10]. Although HamNoSys can be described, annotation is still very manual and therefore large-scale annotation can be both tedious and time-consuming, and can consist of inconsistencies [10][13]. The current computational methods mostly consider individual signs and usually struggle with continuous signing, subtle variations of handshapes and changes of their direction [22][23]. These obstacles create a necessity of automated methods that would be able to work on realistic visual input without violating the structural principles of HamNoSys notation [4][13].

This paper creates an automatic system of producing and authenticating HamNoSys representations of images and video with a particular emphasis on the fundamental phonetic variables of handshape, orientation, location, and movement [10]. Landmark extraction is used to extract hand and body features in the system using MediaPipe [21]. Base configurations and modifiers represent the hierarchical model of handshape recognition, whereas orientation and spatial features are calculated by means of geometric analysis of landmark relationships [15][21]. Location modeling incorporates upper body, head/face, hand, finger, and arm/space position descriptors and movement is described by a two-part analysis that reports major and minor movements [4][10]. Qualitative verification is achieved with the help of an animated avatar that offers an intuitive evaluation of the accuracy of the transcription [25][27].

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
HamNoSys	Hamburg Notation System
SiGML	Signing Gesture Markup Language
ISL	Indian Sign Language
ML	Machine Learning
CSV	Comma-Separated Values

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Sign language is an expressive visual language with many sign masters throughout the world among the deaf and hard-of-hearing societies. Sign languages are unlike spoken languages, which use sound and therefore, the meaning is passed across by affecting the hands, body and facial expressions [16]. Sign languages cannot be generalized, and different regions or nations have their own linguistic scheme, and the differences of the handshape, hand direction, place of movement, and space usage are observed [16][11]. These differences complicate the process of documentation, analysis, and mathematical interpretation.

A significant drawback to the study of sign language is that there is no standardized written language that would be equivalent to the scripts of the spoken languages. Spoken languages also have the advantage of having standardized writing alphabets to capture and process digitally whereas the sign languages do not have a standard format of writing. In order to suppress this drawback, symbolic notation systems like the Hamburg Notation System (HamNoSys) were developed [10]. HamNoSys is a system offering a systematic way of describing signs through encoding phonetic parameters such as handshape, orientation, location and movement [10][13].

Although HamNoSys has descriptive capabilities, the process of transcription is mostly manual, with experts having to annotate data about signings. This is a labour intensive process and time consuming and does not readily scale to large datasets or real-time applications [4][13]. Current computational systems mainly concentrate on single indications and also tend to fail in situations of continuous signing which involve transitions, bi-handed interactions, and dynamic motions [22][23]. Also, there is a serious issue of verification of generated symbolic representations, as most of the approaches do not have trusted guidelines of evaluating transcription accuracy.

This project will overcome these shortcomings by suggesting an automated system of producing HamNoSys syntax on visual input. The model accounts for the information on the major parameters of phonetics through landmark-based feature extraction with rule-directed symbols mapping to create the notation [21][15]. A reconstruction mechanism based on the avatar is presented in order to check the accuracy of generated HamNoSys sequences visually which enhances its reliability [25][27]. The suggested solution will be able to facilitate scalable, interpretable, and effective signing words transcription.

1.2 PROBLEM DEFINITION

Deaf and hard-of-hearing people use sign language as their primary method of communication that requires using a structured arrangement of hand signatures, movements, positioning, and facial expressions. Despite their linguistic richness, sign languages have a fundamental problem; there is no universally standardized written form (similar to that found in spoken languages) of written expression. This is a limitation that makes it difficult to document, disseminate education, and as well as integrate into the computational systems.

HamNoSys provides a systematic symbolic system of describing signs based on the encoding of phonetic features of handshape, orientation, location and movement. Although theoretically successful, the real application of HamNoSys remains largely in the training of manual experts. Human transcription is characterized by frame-by-frame monitoring of signing data and thus high costs of annotating, low scalability, and a risk of inconsistency. These limitations will be a major issue when dealing with large datasets and real-time apps.

Current methods of automation seek to detect features based on image processing or deep learning or skeletal tracking. Most systems have however been developed to support isolated sign recognition and are worse when used in continuous signing. The issues raised in continuous sequences are dynamic transition, coarticulation, orientation variation, and two quick interactions. Also, more articulate characteristics, including flexures in fingers, thumb position, hand position have been simplified or poorly modelled.

A second serious drawback is that there are no known checking procedures of generated symbolic outputs. There are few automated approaches to check the correspondence between these representations and the original signing input even in cases where the HamNoSys codes can be predicted. Such non validation undermines trust in the quality of transcription and limits its usage in a practice.

The issue discussed in this project is hence two pronged. To start with, an automated and scalable approach needs to be developed to produce structural representations of HamNoSys representations of visual sign input especially in continuous signature mode. Second, a mechanism to determine the accuracy of generated symbolic sequences must exist which can be interpreted. This paper suggests a parallel hierarchy in breaking down features of the handshape, location, orientation, movement to allow better symbolic mapping, as well as avatar-driven reconstruction to visually confirm the accuracy of the transcription. The project aims at improving reliability, scalability and usability of HamNoSys based sign language transcription systems by addressing these challenges.

CHAPTER 2

LITERATURE SURVEY

2.1 SURVEY

HamNoSys to Animation / Avatar Generation Systems

This section includes research that focus on changing HamNoSys notation to sign language using avatars. This system mainly focus to imagine symbolic sign representations by producing animated signing movements.

Kaur & Kumar (2016) developed a system that converts HamNoSys notation into SiGML format for Indian Sign Language. The HamNoSys symbols are annotated manually, then they are translated into XML-based SiGML code and this code was used to animate signs using the JA-Signing avatar. They show that symbolic sign descriptions can be changed to visual animations. But, the system works for only isolated signs and needs manual HamNoSys input.

Shalev-Arukshin et al. (2022) made the Ham2Pose model that changes HamNoSys text to 3D skeletal pose sequences. The method uses a transformer based architecture to make real motion from symbolic sign representations. Their idea increase motion accuracy by using a new evaluation metric. But the model depends on correct HamNoSys input.

Sign Language Video Generation and Avatar Synthesis

These works focus on generating real sign language videos or avatars using deep learning models. Instead of symbolic notation these mainly depend on machine learning and neural networks.

Saunders et al. (2020) made SignGAN that adds transformer networks with a mixture network to make real sign language videos. The model produce body poses and uses GAN-based synthesis to make signing sequences. Their results made good continuous signing videos and better than several baseline models. But the approach needs large datasets and high resources.

Forte et al. (2023) proposed SGNify a system that make full body 3D signing avatars from videos. This uses pose estimation with linguistic constraints to give detailed signing motions with hand and facial movements. This makes reality better in avatar generation but it needs high quality training data.

Krishnamurthy et al. (2024) made DiffSign that uses diffusion models to make real sign language videos. The system added 3D avatars with diffusion based motion generation to make coherent signing animations. This allows signer customization and anonymization. But diffusion models need large training datasets and more computation.

Sign Language Translation Systems

This includes research that focuses on changing from spoken or written language to sign language

representations.
Goyal et al. (2020) proposed a grammar based system for changing english lines to Indian Sign Language. The method uses parsing and traditional grammar rules to change words into HamNoSys representations. The HamNoSys that we get in the final is then changed to SiGML for avatar animation. Their system translated complex sentences in a good way but got around a 16% error rate. Because the method depends on rule based grammar, it is hard with ambiguous sentences and unseen vocabulary.

Automatic HamNoSys Annotation and Recognition

These studies focus on predicting HamNoSys annotations from sign language data using machine learning or datasets automatically.
Skobov & Lepage (2020) gave an automated HamNoSys annotation system using machine learning models trained on synthetic sign images. Their idea reduces manual annotation effort by detecting HamNoSys symbols. The method gives a large HamNoSys classification system with many classes. But, the validation accuracy remains low and also the performace for finger movements reduces.

Sign Language Datasets and Benchmark Resources

Some research focuses on making large datasets for sign language motion, annotation, and avatar research. These datasets help train and test sign language models.
Yu et al. (2024) showed SignAvatars, a large dataset having sign language motion data with HamNoSys annotations. The dataset has many videos from many signers doing both individual and continuous signs. This resource make as to be able to research for sign language synthesis and tasks. But the dataset still has less language coverage and needs large scale data collection efforts.
Villa-Monedero et al. (2023) made a Spanish sign language dataset with HamNoSys annotations and 2D pose landmarks. The dataset has many views to support motion generation research. It gives linguistic and motion data that helps improve sign language modeling. But the dataset size is small and only has Spanish sign language.

2.2 SUMMARY OF THE SURVEY AND FINDINGS

Sign language automation has made a huge evolution from a simple symbolic system to the advanced level of Artificial Intelligence. The work by Kaur and Kumar (2016) first focused on conversion of HamNoSys notation to SiGML to animate Indian Sign Language (ISL) [1]. It was limited to animation of individual signs and was not viable for animation of full sentences. Further, the work by Goyal et al. (2020) focused on a rule based approach where the animation of full English sentences could be translated to ISL animation [3]. The rule set was predefined, hence was limited in nature.

Skobov and Lepage (2020) worked on reducing some of the annotation effort required by automatically generating the HamNoSys notation for the sign language video [4]. Interestingly, Shalev-Arkushin et al. (2022) also worked on an approach to translate a symbolized sign language to a human-readable pose estimation [5]. In terms of video fidelity, Saunders et al. (2020) proposed SignGAN to generate highly natural looking signing videos from the input language [2]. Finally, Krishnamurthy et al. (2024) showed that with lots of computational power, DiffSign (a diffusion-based model) can generate very natural-looking and custom signs in any sign language [12].

A few interesting datasets seen. For example, Yu et al. published SignAvatars in the paper "SignAvatars: A Large Scale 3D Motion Dataset for Avatar based ASL Communication" (2023) [20]. More in the "small is beautiful" spirit, research in "Reproducibility in deep learning for AVS: an empirical approach" by Villa-Monedero et al. includes a short description of four small, yet high-quality datasets created to allow the experimentation to be reproducible [18]. A couple of papers highlight the importance of also including linguistic information, to enable the creation of more human, and hence natural, sign avatars: by Forte et al. in "Human-to-Sign Avatar with Realistic Hand Movement through Fusing Linguistic Information" (2023) [7], and by Gil-Martín et al. in "Gloves & Lips: Multimodal Avatar-based Communication for ASL" (2023) [8].

The field is moving toward the development of large scale, data driven, sophisticated models that can manage all the difficulties of human communication. There is still a lot of work to be done to make continuous signing look natural, to capture the subtleties of facial and body language, and to have a dataset that is representative of the diversity of human signers [6][18][20].

2.3 MOTIVATION/CHALLENGES

Less international written form: Sign languages doesn't have written form as spoken languages and documenting, linguistic analysis and computational processing are hard.
Manual HamNoSys drawbacks: Manual HamNoSys annotation has professional specialists and is time taking, inaccurate and cannot be made to large amount of data.

Automated transcription is required: Automation of generation of HamNoSys notation can be of great importance to the efficiency as well as to large scale sign language documentation and accessibility tools.
Poor accessibility to annotated data: Public datasets directly relating sign language gestures to HamNoSys symbols are very less such that dataset preparation can be a big issue.
Hand gesture complexity: Sign language has complex hand signs, contact between fingers, and, in some instances, two-hand signs, which are hard to capture and describe correctly.
Problem with dynamic movements: Dynamic movements and transitional gestures are hard to identify and model in a correct way using a computer vision.

Reliable validation requirement: HamNoSys sequences that are generated automatically are subject to validation so as to make sure that the symbolic representation of the original sign gesture is properly portrayed.

2.4 OBJECTIVES OF THE WORK

The overall goal of the project will be to design and develop an automated structure to generate and validate Hamburg Notation System (HamNoSys) representations, based on an input in a digital visual language, the sign language. The system proposed concentrates on the modeling of core phonetic parameters of the sign language i.e. handshape, orientation, location and movement and at the same time maintaining structural correctness, interpretability and practical usability. The reason for this idea is to break the limitation of the manual HamNoSys process and to give symbolic overview of sign gestures.

The project also needs to go through and test the elements of HamNoSys to understand the symbolic angle of control of hand structures and movement qualities. A good data set is made through the generation of label mappings, CSV based folders of combination that are in order. Based on MediaPipe extraction of landmarks, characteristics are made on hand images to be able to visual gestures in the form of numerical feature vectors to be able to input to machine learning models. To manage the HamNoSys notation, a classification strategy is planned that breaks down gestures to subparts adding base handshape, thumb position, finger bending as well as finger interactions. Also a rule based (with mathematical validation) strategy for location, orientation, movement is implemented.

Special machine learning models are trained to test single attributes of gestures and the results of this are added to give valid combinations of HamNoSys. The algorithms are added to identify and fix invalid outputs with the help of the lexicon based conditions and similarity metrics. Also the avatar based verification pipeline is made through change of generated HamNoSys sequences into SiGML animation that gives a control on comparison between signs and the original inputs. The frame is capable of supporting continuous signing in situations, such as transitional movements, regular scaling, and language independence. The objective of this is to aim at producing reliable system that create and verify a HamNoSys notation of sign language images and videos automatically. This system aims to minimize the manual mapping of the transcription process, achieve high levels of symbolic integrity, and offer an understandable verification system by means of avatars.

CHAPTER 3

PROPOSED WORK

3.1 Architecture of the System

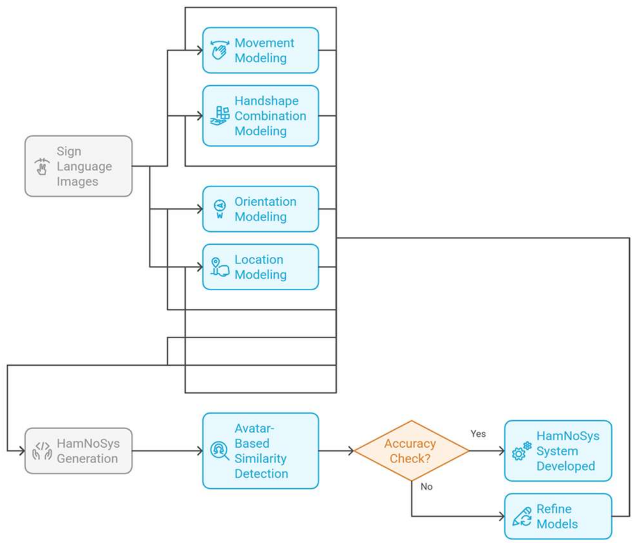


Figure 1: Overall Architecture of the Proposed HamNoSys Generation System

The suggested solution has a modular and hierarchical structure of the automatic creation and validation of Hamburg Notation System (HamNoSys) representations of sign language images and video sequences. Its design supports isolated signs and continuous signing that is done through implicitly combining data preparation, landmark extraction, machine learning prediction, symbolic validation, as well as avatar-based verification into a single processing pipeline.

This system starts receiving visual signs in the form of videos or images. Signs are gathered over publicly accessible datasets and also custom recordings to make them variable and robust. Because there is no explicit dataset that provides HamNoSys annotations, a third-party mapping of the combinations of gestures to the valid HamNoSys symbols is organized. This data will be used as the training and validation one.

The proposed system has a hierarchic design, which provides a video processing system with which HamNoSys notation of sign language videos is rendered automatically. During preprocessing, MediaPipe hands identifies hand landmarks by modelling each hand as 21 three-dimensional keypoints which are both finger joints and palm geometry. It is these landmarks that permit a consistent skeletal description of the hand of the different signers and in dissimilar recording conditions. The conversion of the raw landmark coordinates to geometric and kinematic features follows the feature extraction stage. They include angular measurements between joints, inter-finger distances, orientation vectors, relative spatial positions, and track motions of important data in terms of handshape organization, finger action, spatial relationships, and the dynamics of movement.

The system does not predict the full HamNoSys labels but adopts a hierarchical modeling approach in which the results of the prediction are obtained independently of each other. The handshape module determines the structural arrangement of the hand through a geometric analysis of landmarks in terms of finger articulation and thumb position. The orientation module approximates palm and finger directions by computing vectors based on their geometry of landmarks which allows one to map hand orientation into discrete HamNoSys orientation categories. The location module identifies the anatomical or spatial location of the sign by the analysis of normalized coordinates by reference to facial landmarks, body pose landmarks, and areas of the space surrounding the signer. This module has elaborate sub-components that involve locations of the head and face, locations of finger contact, hand placement in the frame, upper body articulation areas and arm or signing-space location.

There are two complementary modules used to model movement information. The first movement module examines displacement of hand landmarks over time frame between video frames to establish simple motion characteristics of direction, speed, repetition and magnitude of movement. The second movement module is at a higher level of analysis of the trajectory to analyze more important motion attributes like curvy motions, circular motions, oscillations and rotational motions like clockwise or counter clockwise motions. The system uses displacement analysis, path length estimation, turning angle computation, and geometry of trajectory to identify advanced movement modifiers needed in HamNoSys transcription.

All component module generated predictions are united to create candidate HamNoSys representations. These candidates are tested with predetermined symbolic constraints and lexicon-based constraints to check structural correctness and get rid of indefinable combinations of symbols. The sequence which has been validated to HamNoSys is then translated to the format of SiGML that allows one to see the resulting notation in the form of a signing avatar. Lastly, verification via avatars is a comparison between the generated animation and the initial gesture trajectory to measure consistency. The predictions are refined by any inconsistencies that are detected and enhanced system robustness.

3.2 Algorithm Design

3.2.1 Introduction to Algorithm Design

The proposed algorithm will be able to generate valid representations of the Hamburg Notation System (HamNoSys) automatically based on images and video sequences created of sign language. In comparison to the conventional methods that rely on manual transcription or one-stage models of classification, the proposed approach is designed in a modular and hierarchical fashion that enhances the interpretability, scalability and accuracy. The system starts with preprocessing which involves MediaPipe tracking and detecting hand landmarks representing each hand with 21 three dimensional keypoints that describe the geometric structure of the fingers and palm. These landmarks are then converted to significant geometric and kinematic features like joint angles, inter-finger distances, orientation vectors, spatial offsets and motion paths. These aspects give the necessary data needed to examine the configuration of hands, finger articulation, spatial positioning, and dynamics of movement when producing signs.

The special modules that are related to the greater phonetic parts of HamNoSys process these features in algorithm. Handshape is detected with the help of hierarchical machine learning model and forecasts the base handshape, thumb configuration and finger articulation states in a systematic manner. The orientation, location and movement has been estimated on mathematical defined rule based models depending on the landmark geometry and spatial relationship. In the general workflow, the system finds out the spatial location of the hand with respect to the face and body of a signer, the position of the palm and fingers with respect to directional vectors. At this point, the arrangement of the handshapes is identified by a hierarchical division of the locations of fingers and the thumb. Finally, the movement modules refer to what motion direction, speed, repetition and pattern of motion are using trajectories of temporal landmarks. They are predicted fragments which are then put together and validated in view of HamNoSys symbolic constraints to generate structurally appropriate notation results.

3.2.2 Handshape Detection

The Handshape Detection Module is charged with the role of recognizing the hand structure and mapping it to HamNoSys handshape elements. Being a hierarchical modeling system, HamNoSys handshapes are not based on a single label but rather on several attributes, which is why the system is not based on a flat classification model. The organization of the handshape into a few hierarchical nodes that signify various articulatory considerations of the hand. These nodes are the Basic Handshape (A), Thumb Configuration (C), Finger-Type (D), Finger States (E) and optional extra attributes (X) like fingertip or nail contact. The algorithm is able to foretell smaller and more understandable components due to the division of the handshape representation into these nodes that explain the whole hand setup.

On the lowest level of the hierarchy, one may find the Basic Handshape node (A), which determines the general structural shape of the hand. They are fist, flat hand, two-finger shapes, spread finger shapes, pinch form or curved hand shapes. These are further divided into two sub categories namely BASIC-1 and BASIC-2 which are the various structural families of the hand configurations. After identifying the base handshape, the algorithm then attempts to guess the Thumb configuration node (C) that identifies whether the thumb is extended outwards, it is placed across the palm, or in an open position. Once the thumb state is identified, the Finger-Type node (D) looks at the pattern of articulation of the fingers by categorizing the fingers in the form of being straight, bent, hooked, double-bent or being in other forms of bending. The fine articulatory differences between similar handshapes are under this step.

The second rank of the hierarchy is the Finger State node (E), which defines what fingers are in action during the gesture. This involves determining which articulating finger is the dominant, i.e. thumb, index finger, middle finger, ring / pinky, or contact between fingers (between configurations). Other than these important components, the system can also contain of Extra attributes (X), identification of specific points of contact or arthroplastic characteristics, such as fingertip contact, fingernail contact, finger pad contact or finger base or side contact. These properties give a better description was required to trace a fine HamNoSys. To implement such a hierarchical structure, individual machine learning models are implemented at each node. When training, information is bundled in a manner that each individual node is only learned the

names which belong to the specific part. The Support Vector Machine (SVM) node classifiers of nodes A, C, D and E get trained individually on extracted image features.
The prediction of the base handshape with node-A model is also done in the system. According conditionally to this prediction, other models such as the thumb are turned on by the algorithm then finger-type model and finger-state model configuration model. This progressive form of prediction ensures that only the structurally valid paths are made in the hierarchy. The hierarchical prediction of handshape elements in batchy rather than one complex category method is used to even further discrimination in classification less inversion of similar shapes of hands and language congruence and readability on the descriptions produced by HamNoSys, of handshapes.

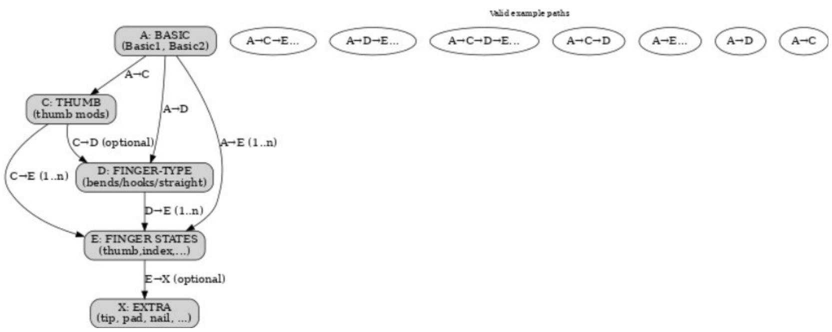


Figure 2 : Hierarchical Structure of HamNoSys Handshape Components

3.2.3 Orientation Detection Module

The directional fit of the palm and fingers is identified as orientation detection which is a major aspect of HamNoSys notation. Orientation is computed by using geometric algorithms of landmark relationships as opposed to direct classification. Orientation has 4 basic parts in it: Bird view, Signer view, Right view, Palm model. If the wrist level is above the eye-level (y-coordinate of wrist is greater than y-coordinate of eye), it is bird view. If the wrist is right side of right shoulder (x-coordinate of wrist greater than x-coordinate of right shoulder), it is right view and else signer view.

The palm axis is the direction that is used to define between the middle finger base and the wrist landmark. A right-hand or directional perpendicular relaxation is lateral orientation. Angular relations between these vectors to the global coordinate system are calculated to provide an idea of palm direction. Normalization of the coordinate representation gives a lack of sensitivity to camera distance and change of signer.

For mapping computed angles to specified HamNoSys orientation actors, Orientation symbols are allocated. Filtering is done at a threshold in order to ignore slight deviations due to landmark noise. This geometric representation offers linear orientation measurement and still is a symbolically interpretable formulation.

Mathematical Validation:
Palm orientation is derived from landmark vectors. The palm axis is defined as:

$$\mathrm{vec}(p) = L_{\mathrm{middle\_base}} - L_{\mathrm{wrist}}$$

A perpendicular direction vector is computed as:
$\hat{n} = (-py\ ,\ px)$

In HamNoSys, these eight finger directions are: right, up-right, up, up-left, left, down-left, down and down-right. Each finger direction occupies one eighth of the 2-dimensional space. 360 degrees of a circle is evenly divided into 8 sections (360/8 = 45) every finger direction has its own section of 45 degrees. This is the only uniform and unbiased partition, with each uniform class spanning ±22.5° about its canonical direction, leading to boundary values of approximately 22.5°, 67.5°, 112.5°, 157.5°, etc. What is implemented and used is the closest integer, 22, 67, 112, etc., which is very close to these values.

3.2.4 Location Detection Module

3.2.4.1. Head and Face Location

In HamNoSys notation, location specifies the anatomical region where a sign is articulated relative to the signer's body. The correct identification of the location that the sign was made is required since a similar hand shape can have different meanings depending on the part of the body that it was made. To make the system work reliably for different body types and for people standing at different positions in the frame normalized landmarks are used instead of raw pixel locations. Such normalization minimizes variability due to camera distance, image resolution as well as personal anatomical variations.

In the case of the head and face location it first identifies the face region by MediaPipe face detection. The identified face bounding box covers the hairline to the chin, which gives a reference frame of geometry that is consistent. The outward facing fingertip or contact point in this area is transformed to a normalized vertical coordinate by the following relation:
$\mathrm{rel}_y = \dfrac{\text{y-face\_top}}{\text{face\_height}}$

where face_height is the distance between the hairline and chin which is vertical. With this, the vertical face positions are brought to the range of [0,1]. They are independent region classification not dependent on face size or camera distance.

It is done by the normalized facial region using the known anthropometric principles of facial proportions partitioned into zones HamNoSys-compatible zones. The positions with the lowest value of less than 0.10 are regarded as satisfying and are explained as hamheadtop, which is the part of the head that is positioned above the skull. The hamhead that includes the skull cap and the hairline area is the interval 0.10 ≤ rely < 0.18. The forehead area that is established in the range of 0.18 ≤ rely < 0.28, and then in eyebrow region in 0.28 ≤ rely < 0.36. Contact points with 0.36 ≤ rely < 0.44 are referred to as hameyes. These thresholds are not empirically tuned but are instead based on normalized facial ratios and therefore the classification is based on natural facial ratios.

It nasal structures are determined by the proportional measurements that are involved in the middle of the face. Points of contact 0.44 ≤ rely < 0.55 and 0.55 ≤ rely < 0.65 respectively give hamnose and hamnostrils. The approximation of the normalization of the position of the nose tip typically applied in anthropometric studies, which extends around the normalization distance of 0.55, is the level around 0.55 the nostril region slightly downwards increases the power to the recognized noise and minor posture variations.

A continuous transition is established between the middle and lower facial parts with lower areas being defined as rely ≥ 0.65. The lips area is characterised in the range of 0.65 ≤ rely < 0.777, the teeth area in 0.777 ≤ rely < 0.888 and the chin area in 0.888 ≤ rely ≤ 1.00. This subdivision is mathematically consistent based on facial proportion theory, in which the lower third of this face is further broken down into functional units that are relatively equal in size. Areas that lie a little lower than the chin are also shaped: those points in which the facial edge is touched are referred to as hamunderchin, those far lower as hamneck. This extension enables the signs that are articulated in the lower head or neck to be recognized accurately.

Besides vertical segmentation, horizontal partitioning is employed in differentiating lateral facial parts. The horizontal coordinate is normalized with respect to the face bounding box using the following relation:

$$\mathrm{rel}_x = \frac{\text{y-face\_left}}{\text{face\_width}}$$

This transformation converts horizontal coordinates to [0,1], which is an invariance to changes in face size and viewpoints of a camera. According to the classical theory of facial proportions, five equal parts were split on the horizontal axis. Ranges of $0.00 \leq \mathrm{rel}_x < 0.20$ and $0.80 \leq \mathrm{rel}_x \leq 1.00$ are the areas of the ear that are lateral and would be considered hamear or hamearlobe. Middle ranges $0.20 \leq \mathrm{rel}_x < 0.40$ and $0.60 \leq \mathrm{rel}_x < 0.80$ are the cheek areas and they are marked hamcheek. The key major midline facial features that include the nose, lips, and chin are located in the central area $0.40 \leq \mathrm{rel}_x < 0.60$. The given symmetric partitioning guarantees consistent classification of the lateral facial areas in a manner that is independent of the size of an image and variations between subjects.

The thresholds and segmentation approach in this module is based on the known anthropometric research studies on facial proportions, so that the location classification achieved does not depend on dataset-specific heuristics.

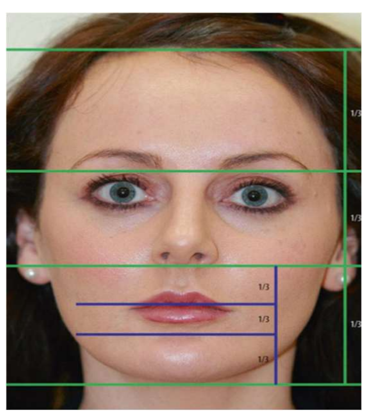


Figure 3: Facial Proportion Segmentation for Head and Face Location Detection

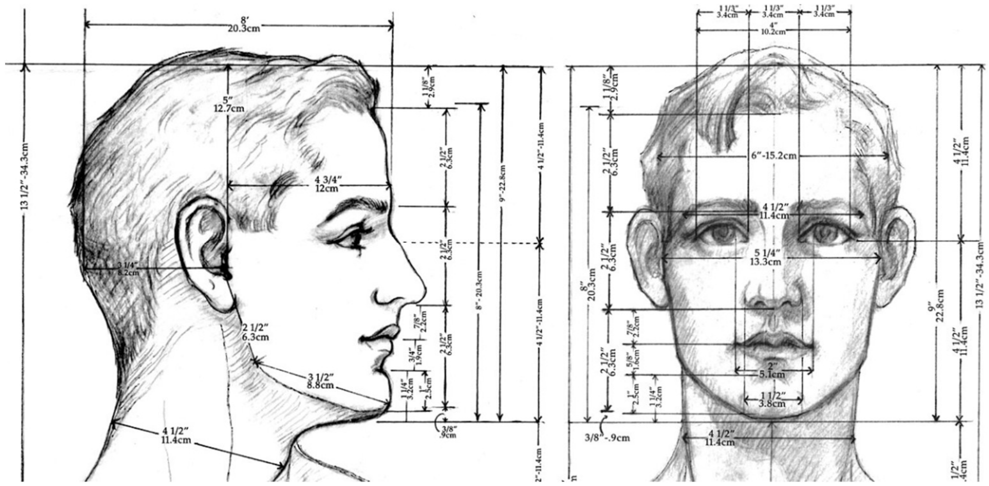


Figure 4: Anthropometric Reference of Human Head and Facial Measurements

It is a diagram of anthropometric values of the face and head of a human being that are used as reference values to calculate normalized spatial ratios. It is these body sizes that assist in computing geometric thresholds on which the head and face location module operates[31].

3.2.4.2 Contact Types

Contact types are employed to characterize the way of how the two articulators (or one of the fingertips) come into physical contact with each other. Touching any of the hands, and objects or other parts of the body. The geometric is used to identify these interactions. Normative MediaPipe landmark locations were used to obtain values. Since all coordinates are normalised (between [0,1]) where spatial thresholds are proportional distances. The system can be robust with various cameras in respect to the size of the body of the signer distances and differences of subjects.

Hamtouch: The hamtouch name is used when the Euclidean distance between two contact points is very low. Assuming that p1 and p2 are the positions of two landmarks in normalized coordinates, where p1=p2. Contact distance is computed to be $d = \|p_1 - p_2\|$. In the case of $d \leq \epsilon$, the points are said to be touching.

The threshold $\epsilon = 0.02$ that is set is based on anatomical proportions. MediaPipe coordinates are taken to be normalized, where 1.0 is the approximate body height. Having the mean human height of 170 cm and the mean width of fingers of 1.5-2 cm, the proportion of it normalized to the mean human height of 170 is $2/170 \approx 0.012$. A slightly bigger value of tolerance is thus selected which considers curve of fingers and slight tracking noises. This will make sure that real physical contact is identified easily**.**

Hamclose: When two points are very close to each other but are not touching, it is called as hamclose label. They are noted by $\varepsilon < d \leq 0.05$. Given that 0.05 is 5% of normal body height, this distance represents that a distance of 8 cm (0.05 * 170 cm). This distance means very close but clearly a not in direct contact (touch).

Hambrushing: When the frames are moving, and if theree are two points close to each other out of several frames, then the hambrushing label finds them out. Both the distance factor and an active motion are needed. Equation of Motion Magnitude is: $\Delta p = ||pt+1 - pt||$.
Whenever $\Delta p > 0.01$ and the two points are within the Hamclose range, then the motion will be taken brushing. The value of 0.01 means about 1-2 cm of movement in the physical world and is enough to be differently categorized as wantedly moved vs. slight landmark jitter.

Hambehind: The front-back relationship of two points in physical space, which is given by the depth (z) coordinate is shown by the hambehind label. Equation of the distance relationship is: $||z1 - z2|| > 0.03$. If this condition exists, then one articulator will be considered to be behind the other. The average depth (30-35 cm) of the human torso represents approximately 1 cm of actual distance and this will be used as the threshold to differentiate actual depth differences vs. depth noise..

Hamcross: When one hand crosses over the another hand, it detects the hamcross configuration by going throgh the horizontal placement of both hands over time. It is also done by seeing the relative horizontal positions of both the hands when the signs change.
$(W_{L,x} - W_{R,x}) \times (W_{L,x'} - W_{R,x'}) < 0$
The difference in depth verifies that there is spatial overlap and the motion is classified as a crossing. Depth check is used to ignore false positives. This happens due to the simple overlap in the image plane that takes place.

Haminterlock: When multiple pairs of fingers are very close to each other but they are located at different depths, the haminterlock condition occurs. Several pairs of fingertips are counted by the system. The system also counts how many of the counted pairs actually meet the needed condition. If multiple finger pairs satisfy both the required distance and the difference in depth then the configuration will be classified as interlocking.

3.2.4.3 Finger Location Detection

Finger location detection tells which finger out of all is active and also says which part of the finger is occupied.Landmark indices for each finger are fixed in MediaPipe which proves the ability to map landmarks to anatomical structures in a consistent way.
The active finger is determined by examining geometric features of the active finger such as finger extension relative to the palm, distances between instrumental points and the location of center of hand and joint (i.e., the three finger joints). Using the fingertip landmarks and the center point of the hand, the finger extenstion can be counted as shown below: $d_i = \|p_{tipi} - p_{palm}\|$
The articulator will be the finger that shows the maximum amount of finger extension as compared to its expected length.

Hamthumb: The hamthumb classification is used in the instance when the thumb is being. When the thumb is being used as the primary articulator, it will occupy landmark 1 through landmark 4 in the MediaPipe landmark index and the algorithm will analyze thumb extension and thumb joint angle with respect to the palm. When a thumb is extended or active in contact, the algorithm indicates that the hamthumb is the current thumb configuration.

Hamindexfinger: Only when the index finger (landmarks 5 to 8) is the active finger , the classification of Hamindexfinger is possible. It is also important that it should have highest amount of extension from the other four fingers. In order to determine how much greater the extension is for the index finger than that of the other fingers, we will compare the lengths of both (1) the distance from tip of the index finger to the center of the palm, and (2) the degree of which the index finger is either completely straight or fully bent in relation to other fingers.

Hammiddlefinger: The hammiddlefinger classification indicates when a person's middle finger (landmarks 9-12) extends the most from the center of their palm in measurement. The middle finger usually extends the most of any finger; therefore, it normally will extend the most from the center of the palm relative to the other fingers middle finger's extension outwards from the center of the hand is calculated by the hammiddlefinger classification. It is done based on a line drawn from the center of the hand to the fingertips of all the fingers. Typically, the middle finger has the greatest amount of extension of all fingers on a hand; thus, it also tends to extend the farthest away from its center relative to other fingers.

Hamringfinger: The hamringfinger classification identifies whether or not the ring finger (landmarks 13-16) is the dominant finger of articulation(i.e., how much it can extend and/or interact with its respective palm compared to other fingers).

Hampinky: The hampinky system identifies a little finger as the dominant articulated finger(land marker numbers 17 - 20). The hampinky can be distinguished by measuring the distance between the palm and the endpoint of the little each other finger.

Finger Contact Region Classification
In addition to identifying which finger is active, the system determines which part of the finger participates in the contact.

Hamfingertip: The hamfingertip is the contact point closest to the fingertip landmarks. This takes place at the minimum Euclidean distance between a contact point and a fingertip.

Hamfingermidjoint: The hamfingermidjoint is the classification for the contact point closest to mid-joint landmarks. This happens with a bent finger where the finger joint is in contact with another surface than the fingertip.

Hambetween: The hambetween contact point is approximately equal distance between two adjacent fingers. The object/contact area is therefore located between two fingers as opposed to directly contacting one.

Hamfingerpad: The hamfingerpad is a contact point located on the palmar-facing surface of the fingertip. This is determined by the relative orientation of the fingertip and the palm.

Hamfingernail: The contact point occurs only on the nail side of the fingertip. This is said by hamfingernail. The vector is compared from the fingertips to palm's center to the vector from fingertips to contact point.

3.2.4.4 Hand Location Detection

The location of the hand is detected to identify the spatial area that the hand is in the frame. Such information is useful in determining the location of gestures with reference to camera point of view. The system works with normalized image coordinates also to get resistance to changes in camera resolution and image size, with every identified landmark location denoted as: P = (x,y), x,y ∈ [0,1].

These normalized coordinates are used to make sure that the spatial classification is not dependent on video resolution and scaling. The centroid of the identified hand landmarks is calculated to give a hand representation of the overall position of the hand. Centroid is a representation that gives a constant location of the hand in the frame.

Let P_1,P_2,...,P_n denote the detected hand landmarks. The centroid C is defined as

- $C = (C_x, C_y)$
- $C_x = \frac{1}{n}\sum_{i=1}^{n} xi \, , Cy = \frac{1}{n}\sum_{i=1}^{n} yi$

The centroid provides a stable representation of the hand's spatial location in the frame.

Horizontal Hand Position Classification

To establish the position of the hand, the horizontal axis of the normalized coordinate space [0,1] is subdivided into three equal parts to establish whether the hand is on the left, center, or right side of the frame.

handleft - Hand on Left Region.
Condition: $C_x < 0.33$
Explanation: The values below 0.33 display, when the centroid lies within this range, the hand is on the left side.

handcenter - Hand Located in Center Region
Condition: $0.33 \leq C_x \leq 0.66$
Explanation: This time frame represents the main part of the frame, where captures all gestures that are made in the general center of the frame.

handright - Hand On Right Region.
Condition: $C_x > 0.66$
Explanation: The values above 0.66 display the final part of the frame, which means that the hand is on the right side.

Vertical Hand Position Classification

In the same way, the vertical axis is subdivided into three parts in order to determine whether the hand is in the lower, upper, or central position of the frame.

handup - Hand in upper part
Condition: $C_y < 0.33$
Explanation: The values below 0.33, which means that the hand is on the upper side.

handdown - Hand in lower part
Condition**:** $C_y > 0.66$
Explanation: The values above 0.66, which means that the hand is on the bottom side.

Mathematical Validation of Hand Location Classification

Euclidean Distance Validation

Euclidean distance between points can be calculated to check the spatial consistency of the hand landmarks that have been detected:

$$d(A,B) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

and $A(x_1,y_1)$ and $B(x_2,y_2)$ are coordinates of landmarks. This measure of distance is used to make sure that the landmarks identified have realistic spatial associations.

Horizontal Offset

The horizontal movement of the hand centroid relative to the starting frame is: $x_{offset} = C_x$

This value is what defines the hand lateral position in the frame.

Vertical Offset

On the same note, the vertical displacement will then be calculated as: $y_{offset} = C_y$

This value determines the location of the hand in the frame vertically.

Majority Voting Validation

Because the hand position can vary slightly between frames because of the noise in the detection, it is classified with the help of a majority voting strategy. When frame-level predictors are: $P=\{L_1,L_2,L_3,\ldots,L_n\}$

The last place of location is calculated as:: $L_{final} = mode(P)$

Using the label that appears most often across frames helps reduce tracking mistakes and makes the classification more consistent.

Scale Invariance Justification

Everything is computed in normalized co-ordinates: $x,y \in [0,1]$

This makes sure that the classification process is not dependent on camera resolution, size of the image or video magnification. As a result, the hand location detection module is clearly consistent irrespective of recording environments and devices.

3.2.4.5 Upper Body Location Detection

Location detection of the upper body location determines the location of the hand contact with the anatomic regions of the upper part of the body. These points are some of the significant articulation points in HamNoSys notation; the shoulder, elbow, arms and the torso. The proposed system identifies these points based on skeletal features, which are based on pose estimation. As landmark coordinates are normalised in the range [0,1], the spatial reasoning can be done in a scale-independent way.

The upper body locations which are in this system are: hamshouldertop (contact above shoulder), hamshoulders (contact near shoulder line), hamelbowinside (contact near elbow joint) and hamupperarm (contact along the upper arm), hamlowerarm (contact along forearm), hamwristback (contact near the wrist) and hamchest (contact on upper torso), hambelowstomach (contact on lower torso).

Mathematical Validation of Upper Body Location Classification

Relative skeletal geometry is based on pose landmarks, which are used to determine upper body locations. Every body point is notated using normalized coordinates:
$P = (x,y),\ x,y \in [0,1]$

Landmarks used as references:
- Left Shoulder → LS
- Right Shoulder → RS
- Left Elbow → LE
- Right Elbow → RE
- Left Wrist → LWL
- Right Wrist → RW
- Left Hip → LH
- Right Hip → RH

The hand contact point is estimated as the centroid of detected hand landmarks: $C = (C_x,C_y)$
The torso center is defined as the midpoint between the shoulders: $T_x = \frac{LSx + RSx}{2}$

Two body reference scales are computed to normalize spatial measurements:
Shoulder width: shoulder_width = d(LS,RS)
Torso height: torso_height = $hip_y - shoulder_y$
where $hip_y = \frac{LH \quad + RHy}{2}$, $shoulder_y = \frac{LS \quad + RS}{2}$

1. Normalized Distance Measure
Spatial proximity between two body points is measured using Euclidean distance.

$$d(A,B) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

To ensure scale invariance, the distance is normalized using a body reference length:
$d_{norm} = \frac{d(A,B)}{scale}$

where the scale may correspond to shoulder width, arm length, or torso height.

2. Shoulder Proximity Detection
The shoulder region is identified using the normalized distance between the contact point and shoulder landmarks.

Condition: $d_{norm}(C,LS) < 0.25$ or $d_{norm}(C,RS) < 0.25$

Classification rule:
If $C_y < shoulder_y$ → hamshouldertop
Otherwise → hamshoulders

Explanation: Distances within approximately 20–25% of the shoulder width correspond to locations near the shoulder joint.

3. Elbow Proximity Detection

The elbow region is detected using the normalized distance between the hand contact point and elbow landmarks.
Upper arm length is computed as upper_arm_length = d(Shoulder,Elbow)
Condition: dnorm(C,Elbow) < 0.20
Classification: hamelbowinside
Explanation: At a threshold of approximately 20% of upper-arm length gives contact positions near the elbow joint.

4. Arm Segment Classification
Vertical relationships amongst shoulder, elbow, and wrist are examined and the displacement laterally relative to the torso, to differentiate between the location of upper arms and forearms.

Lateral offset from the body center: lateral_offset = $|C_x - T_x|$
Condition for lateral separation: lateral_offset > 0.25 × shoulder_width
Condition of the upper arm: $\text{shoulder}_y < C_y < \text{elbow}_y$
Classification: hamupperarm
Condition of the lower arm: $\text{elbow}_y < C_y < \text{wrist}_y$
Classification: hamlowerarm
Explanation**:** Big side movement is an external position of the hand, at the side of the torso silhouette, which is an arm area, as opposed to a torso area.

5. Wrist Back Detection
The detection of contact between the wrists is done through the normalized distances based on forearm length.

Forearm length: forearm_length = d(Elbow,Wrist)
Conditions: $d_{norm}(C,\text{Wrist}) < 0.10$ and $d_{norm}(C,\text{Elbow}) > 0.40$
Classification: hamwristback
Explanation: The close proximity to the wrist combined with a further distance to the elbow will assure that the point of contact will be concentrated around the wrist.

6. Torso Region Classification
In case the point of contact is on the vertical (between the shoulder and the hip level), in that case the torso parts are defined on a normalized vertical ratio.

$$\text{rel} = \frac{\text{Cy} - \text{shouldery}}{\text{torso\_height}}$$

Classification rules:

- rel < 0.35 → hamchestrel
- 0.35 ≤ rel < 0.70 → hamstomach
- rel ≥ 0.70 → hambelowstomach

Explanation: The use of the proportional dividing of the torso height to give a scale independent means of identifying the regions of the chest, stomach, and lower torso.

Scale Invariance Justification
The nature of scale invariance is that an object in the world has a collection of geometric properties which do not change as a function of its size/scale. Every spatial threshold is calculated with respect to normalized distances with respect to anatomical reference lengths:

$d_{norm} = \frac{\text{distance}}{\text{body scale}}$

with body scale being equal to upper arm length, shoulder width or torso height. Such normalization makes the classification procedure immune to the changes in camera distance, body size, image resolution, and video scaling.

3.2.4.6 Arm & Space Location

hamdoublebent - Both arms bended
Condition: $\theta_L < 120°$, $\theta_R < 120°$
Proof: Normal elbow articulation of humans is 30+ degrees to 160 degrees. It is also biomechanically consistent since the curved positions are less than 120 cluster.

hamarmextended - Fully extended arm
Condition: $\theta > 160°$, $R > 0.95$
Proof: One of the limbs is almost at right angles. The extension ratio is discovered to be near to one in affirming numerous proximities of the wrist to maximal reach.

hamlrat - Hand at body midline
Condition: $x_{offset} < 0.04$
Proof: Shoulder width ≈ 0.35–0.40 (normalized). The scale is also invarient to midline tolerance which is a small proportion of half-shoulder span.

hamlrbeside - Hand next to body
Condition: $x_{offset} > 0.10$, $z_{offset} < 0.08$
Proof: The lateral shift of the movement of body sides is gigantic and the forward movement is small. The mix up with neutral signing space is avoided by depth constraint.

hamneutralspace - Signing space without hamneutralization.
Condition: $z_{offset} > 0.05$
Proof: z normalized is a good representation of forward movement. The plane of the torso and volume of signing space is differentiated by existence.

Mathematical Validation of Arm & Space Location Classification
The MediaPipe Pose landmarks give relative skeletal geometry that is used to get the position of the arms and the signing-space. Every landmark is notated using normalized coordinates in 3D plane:
$P = (x,y,z),\ x,y,z \in [0,1]$

Reference landmarks:
Shoulder → S. Elbow → E, Wrist → W, Torso center → T (midshoulders)

1. Elbow Angle (Arm State Descriptor)
The angle of the joint: $\angle(SEW)$ is the measurement of the bending of the arm.
Computed via cosine rule: $\theta = \cos^{-1}\left(\frac{(S-E)\cdot(W-E)}{\|S-E\|\ \|W-E\|}\right)$
Interpretation: Small angles → bent arm and Large angles → extended arm

2. Arm Extension Ratio

Arm extension is scale-invariant: R = $\frac{\|S-W\|}{\|S-E\|+\|E-W\|}$

For a fully extended arm: R → 1

3. Horizontal Body Offset

Relative lateral displacement: $x_{offset} = |W_x - T_x|$

No need to use absolute pixel position.

4. Depth Offset (Signing Space Indicator)

Forward/backward displacement: $z_{offset} = |W_z - T_z|$

Brings the difference between the signing space and the plane of the torso.

Scale Invariance Justification:

All the thresholds are normalized: x,y,z ∈ [0,1]

Thus, it is not categorized by the distance of the camera, body and resolution size.

3.2.6 Movement Detection Module

Movement 1

The Movement 1 module designed is to recognise and determine the significant movement characteristics of the hand anytime it is producing a sign. Representation of movement in HamNoSys-based transcription is not portrayed as a single categorical name but as a structured form of representation of many attributes of movement including direction, depth, speed, repetition as well size of the movement. To enable a systemic cognition of the motion, the module is executed using the hand movement as a time-varying sequence of three-dimensional landmark locations determined out of video frames. Mathematically the path of the hand is given by:

$p_t = (x_t, y_t, z_t)$

where t denotes the frame index, and (x_t, y_t, z_t) are normalized coordinates in the range [0,1], following the MediaPipe landmark convention. Frame to frame motion is quantified in frame to frame motion: $\Delta p_t = p_{t+1} - p_t$

This displacement at one point in time is identified in this displacement as a momentary spaceward and inward-looking vector. In order to estimate the overall trend in movement direction in the entire gesture, the module averages the movement displacement vector across N frames: $(\underline{d}_x, \underline{d}_y, \underline{d}_z) = \frac{1}{N-1}\sum_{t=1}^{N-1} p_{t+1} - p_t$

Noise Suppression and Dead-Zone Threshold:

Tracking noise can easily create small noises since landmark coordinates are normalized meaning there isn't necessarily a movement taking place. To prevent the perception of such jitter as motion, a dead-zone threshold is added: ε = 0.02

This value is equated to a frame span of approximately 2% amongst the normalized frame span, i.e. any displacement value that is smaller than the same is regarded as noise. Thus, only changes that are statistically significant are considered in the classification.

Direction Classification

Direction of movement is derived by taking into account the constituents of mean displacement register with respect to the threshold ε. The -axis element leads to the interpretation of the horizontal movement. A hand is said to be in motion (hammover) when $(d_x > \varepsilon)$ and is said to be in motion (hammovel) when $(d_x < -\varepsilon)$.

Vertical motion is determined with the help of the -axis. The image coordinate rotation in this case is that which makes $(d_y < -\varepsilon)$ the upward movement (hammoveu) and $(d_y > \varepsilon)$ the downward movement (hammoved). To estimate the depth movement, z-axis component is applied. The negative displacement surpassing the threshold $(d_z < -\varepsilon)$ indicates the motion towards the body (hammovei) and the positive displacement $(d_z > \varepsilon)$ indicates the motion away (hammoveo).

When more than one component of the displacement exceeds thresholds simultaneously, this is when we have compound directions. In one instance, hammoveil and hammoveor are hammoveil and hammoveor respectively caused by depth inward and depth outward movement respectively. This multi axis analysis offers appropriate recording of complex spatial movement.

Speed Estimation

Speed is quantified using the magnitude of instantaneous displacement vectors:

$$v_t = \|\Delta p_t\|$$

The module computes the mean per-frame speed:

$$v = \frac{1}{N-1} \sum_{t=1}^{N-1} \|\Delta p_t\|$$

Standardized scores have managerial-grounded thresholds. If v < 0.01 is considered to be slow and monitored (hamslow). The motion is known as fast (hamfast) when v > 0.03. It has an approximation of 1% and 3% normalized displacement per frame respectively.

Repetition Detection

The repetition patterns are established depending on whether the change in velocity is of predominance, or not. Instantaneous velocity is defined as:

$$v_t = x_{t+1} - x_t$$

Direction indicator:

$$s_t = sign(v_t)$$

A direction reversal is the one where:

$$s_t \cdot s_{t+1} < 0$$

The type of repetition depends on the quantity of such reversals. A single change in direction means hamrepeatreverse, a succession of changes in direction means hamrepeatcontinue and repetitions means hamrepeatseveral. The process measures the cyclic movement despite the magnitude of absolute movement.

Movement Size Classification

Movement size is derived from the spatial extent of the trajectory:

$$A = max(p_t) - min(p_t)$$

If A<0.02, then this is small (hamsmallmod). The tag associated with a large (hamlargemod) motion is A.08. The dead-zone values are implicitly hamnomotion which is in agreement with rules being suppressed by noise.
All classifications follow a unified threshold-based rule:

- $|measurement| > threshold \ \rightarrow \ valid\ movement$
- $|measurement| \leq threshold \ \rightarrow \ noise/ignored$

Movement 2

Movement-2 module is another dynamically interpreting scheme of hand movement that models higher order features of the path, as opposed to simply displacement. whereas Movement-1 possesses the most important attributes of direction and speed, the Movement-2 is situated on geometric and kinematic attributes as curvature, circularity, rotational way, oscillatory conduct and path form. They have acquired these properties by a mathematical investigation of landmark trajectories, displacement vectors, the turning angles and the relationships of displacement and path-length. These results show that according to the system, movement modifiers can be predicted according to the situation like circular movement, diagonal curves, clockwise rotation or anti-clockwise rotation, stirring and repetitive oscillation. The resultant Movement-2 predictions are descriptive motion data that is complementary to Movement-1 generated data and then harnessed with HamNoSys output pipeline. Stratification will be used to make sure that the fine-grained structures of movement that are needed are being well captured without affecting the symbolic validity.

Mathematical validation of Movement 2 (Trajectory and path modeling)

Movement 2 models path shape, curvature, circularity, and rotational manner, independent of basic direction/speed.

1. Trajectory Representation

Hand motion across frames: $T=\{p_1,p_2,...,p_N\}$, $p_i \in R_3$

2. Displacement vs Path Length

Net displacement: $D = \|p_N - p_1\|$

Total path length: $L = \sum_{i=1}^{N-1} \| pi + 1 - pi \|$

Circular / Looped Motion Criterion: $D \ll L$

Proof:

For linear motion: $D \approx L$, For circular motion: $D \rightarrow 0$, while $L > 0$

3. Turning Angle (Curvature Measure)

Local trajectory curvature: $\theta_i = \cos^{-1}\left(\frac{vi \cdot vi+1}{\|vi\|\ \|vi+\ \ \|}\right)$

Mean turning angle: $\bar{\theta} > \varepsilon$

Large values → curved motion.

4. Plane Detection

Variance along axes determines motion plane:

- XY plane → planar circular,
- XZ plane → depth circular,
- YZ plane → vertical-depth circular

5. Clockwise vs Counter-Clockwise Proof

Angular position relative to centroid: $\alpha_i = \tan^{-1}\left(\frac{yi-\bar{y}}{xi-\bar{x}}\right)$

Rotation direction: $d\alpha/dt < 0 \Rightarrow$ Clockwise

6. Stirring / Rotational Manner

Condition: $D \approx 0$, $L > 0$, $\bar{\theta} > \varepsilon$

Meaning → rotation without translation.

7. Oscillation Detection

Sign changes along dominant axis: $\mathrm{sign}(vi,k) \neq \mathrm{sign}(vi+1,k)$

Repeated reversals → repetitive motion.

3.3 DATASET FOR STUDY AND PLATFORM

3.3.1 Dataset

The format of the dataset used to train and test the implemented system is very important to the performance of the system. The lack of public available datasets for sign images or videos with HamNoSys symbolic annotations is one of the biggest problems with HamNoSys-based sign language transcription. To avoid this drawback a special dataset was made in reference to the handshape module. The dataset was sformated according to the principles of HamNoSys handshape notation so that every class is an allowed combination of base handshape, thumb arrangement, finger interaction pattern. 149 different combinations of handshapes were characterized based on HamNoSys principles. The combinations were saved in a hierarchical structure by folders and a naming pattern depending on its HamNoSys descriptor directly.

First, one base image was collected by hand per handshape category and then methods of data augmentation were used to make the data more various. Among the augmentation operations were rotation (±20°), scaling, horizontal flipping, translation, brightness manipulations and noise addition. In this way, each of the classes was multiplied by 21 variants, forming a collection of more than 3,000 images. These augmentations are used to simulate changes in hand orientation, lighting, and camera perspective and enhance the robustness of the trained models as well as their generalization. In order to have some symbolic consistency, a CSV mapping table was developed which correlates each combination of gestures with a HamNoSys command and hexadecimal value. The ground truth used in the training is this mapping file, it can also be used to validate the predicted combinations of symbols and also the HamNoSys sequences generated can be guaranteed to be of valid symbolic structures.

Although a dataset was developed in the handshape detection module, it is much more challenging to develop annotated datasets in terms of orientation, location and movement components since these attributes need the use of continuous video sequences and annotation frame by frame. Thus, the rule-based approaches based on the MediaPipe landmarks and mathematical equations are used to implement these modules, but not machine learning models. Orientation, location and movement are computed respectively with palm direction vectors and angular thresholds, normalized spatial relationships between hand and body landmarks and temporal changes in landmark trajectories across frames respectively. Such a combination of custom handshape dataset, symbolic CSV mapping and mathematically defined rule-based modules offers a viable and stable structure to a practical approach to automated HamNoSys generation.

3.3.2 Platform and Specification

Platform Used

Google Colab Pro

The development, training, and experimentation for the proposed system were carried out using Google Colab Pro, a cloud-based computational environment that provides GPU acceleration, extended runtime sessions, and high-memory virtual machines. This platform allows us to execution of the complete pipeline along with MediaPipe based landmark extraction, feature engineering, hierarchical model training, and evaluation. The availability of GPU resources accelerated feature computation and model training also extended runtimes allowed without any breaks processing of large datasets and experiments without depending on local hardware.

HamNoSys Reference and Avatar Tools

With computational platform many special online tools and resources were used during development and testing of HamNoSys notation:

1.HamNoSys Input Tool

This tool was used as a reference interface to test the correctness of the output HamNoSys symbols and their corresponding representations. It gave an official reference for making and testing HamNoSys notation.

2.JASigning / Avatar Animation Client
This platform was used to change HamNoSys based SiGML scripts to avatar performances. It allowed verification of the output notation by giving the sign language movements.

3.LinguaSign Sign Files Repository
This repository gave sample SiGML sign files used for testing avatar and understanding the mapping between HamNoSys notation and animated sign execution.
These tools were important for testing the generated HamNoSys sequences and make sure that the symbolic representations corresponded to the sign gestures.

Software Stack

The proposed system is deployed on basis of the combination of computer vision, machine learning, and data processing libraries.

MediaPipe: This is applied in hand landmark detection and tracking. MediaPipe identifies 21 hand landmarks in each hand and gives normalized coordinates of three dimensional points, which are the basic features for gesture recognition

OpenCV (cv2): It is used in the steps of preprocessing images like resizing, frame extraction, frame augmentation, and visualization.

Scikit-learn: It is used in the steps of preprocessing images like resizing, frame extraction, frame augmentation, and visualization (landmark detection).

XGBoost: It is applied as the main machine learning algorithm to classify components of a handshape hierarchically. It is used as a default global classifier in order to solve ambiguous HamNoSys combinations because it is strong in addressing high-dimensional feature space.

Pandas: Used in the manipulation of CSV mapping files, dataset management and structured data.

NumPy: Applied to the computation of numericals based on landmark coordinates, geometrical feature extraction, and calculations on vectors.

Matplotlib / Seaborn: Used for visualization of dataset distributions, training performance, and evaluation metrics.

Joblib: Used for efficient serialization, storage and loading of trained machine learning models.

CHAPTER 4

RESULTS AND INFERENCES

4.1 METRICS FOR EVALUATION

In order to determine whether the created HamNoSys representations are correct, a verification method is applied through an avatar. Once HamNoSys notation of a particular sign has been created it is translated into SiGML and displayed via a signing avatar. The generated avatar animation is graphically compared to the input gesture to determine whether generated HamNoSys sequence recreates the sign. Cosine similarity is also employed to determine the similarity between the landmark features of the input gesture and the avatar-generated gesture besides visual inspection. The greater cosine similarity value shows that the reconstructed avatar movement is closely similar to the original signing pattern which proves the validity and accuracy of the generated HamNoSys notation.

4.2 PARAMETERS SETTINGS

The proposed system uses mathematically defined parameters and threshold ranges based on normalized MediaPipe landmarks to compute spatial and motion features required for HamNoSys generation. For head and face location detection, MediaPipe face detection provides a stable bounding box from the hairline to the chin. Instead of pixel coordinates, normalized coordinates are used to ensure robustness against camera distance and signer variations. A relative vertical coordinate is computed as:

$$\text{rel\_x} = \frac{\text{x} - \text{face\_left}}{\text{face\_width}}$$

This value determines the position of the fingertip within the face region. Based on anthropometric facial proportions, the normalized facial region is divided into zones corresponding to HamNoSys facial locations such as head top, forehead, eyebrow, eye level, nose, nostrils, lips, teeth, chin, and neck. This divides the face into symmetrical regions representing ear, cheek, and central facial structures, enabling scale-independent classification of facial articulation locations.

To determine contact types and finger locations, spatial relationships between hand landmarks are analyzed using Euclidean distance, cosine similarity, and depth comparisons. Threshold values are defined based on normalized body proportions. For example, two points are classified as touch (hamtouch) when the distance between them is approximately equal to the normalized finger width ratio (around 0.012 of body height). If the distance is slightly larger but still within a defined range, the contact is classified as close (hamclose). When points remain very close while the hand is moving across multiple frames, the interaction is classified as brushing contact (hambrushing). Additional geometric reasoning is used to detect behind, crossing, and interlocking finger configurations by analyzing relative depth values and positional changes. Active finger roles such as thumb, index, middle, ring, and pinky are determined based on finger extension, straightness, and distance from the palm. Contact surfaces such as fingertip, fingernail, finger pad, and mid-joint are identified using minimal distance comparisons between relevant landmarks.

For hand and body location classification, the hand centroid is computed from detected hand landmarks and represented as: $C = (C_x, C_y)$

The normalized frame is divided horizontally into left, center, and right regions, and vertically into upper and lower regions to determine the general hand position. To improve stability, majority voting across frames is used to reduce temporary detection noise. Upper body articulation locations such as shoulder, elbow, wrist, chest, stomach, and lower torso are determined by measuring normalized distances between the hand position and body landmarks obtained from MediaPipe Pose. These distances are scaled using anatomical references such as shoulder width, arm length, and torso height, ensuring scale invariance across different body sizes and camera perspectives.

The movement detection module analyzes temporal landmark trajectories to extract dynamic motion characteristics. In Movement-1, frame-to-frame displacement vectors are calculated to derive movement direction, speed, repetition patterns, and movement magnitude. To filter tracking noise and overlook small variations small dead-zone threshold ($\varepsilon = 0.02$) is applied .The higher-level motion patterns of curved paths, circular motion, oscillation and rotating behavior are studied, as geometric structure of the trajectory is studied in Movement-2. \\This is done through the comparison of the net displacement with the total path length, calculation of the turning angles between the displacement vectors and analysis of angular change with the centroid of a trajectory. With a combination of these parameter settings, the system is capable of effectively capturing basic directional movements as well as complex motion trajectories and, therefore, represents the motion effectively in the HamNoSys transcription pipeline.

4.3 RESULTS AND DISCUSSION

The system that was proposed produces HamNoSys notation by examining some of the major elements of sign articulation, such as location, orientation, handshape, and movement.

4.3.1 HandShape

This figure shows a sample input hand gesture used for handshape recognition. The hierarchical handshape model analyzes the extracted hand landmarks and predicts the corresponding HamNoSys handshape combination. In this example, the system correctly identifies the handshape as *hamcee12, hamfingerstraightmod, hamindexfinger,* and *hammiddlefinger*.

Figure 5: Sample input and predicted handshape

Avatar Verification

To determine whether the prediction of HamNoSys representation was accurate, the generated avatar gesture is compared with the input hand gesture based on the cosine similarity.

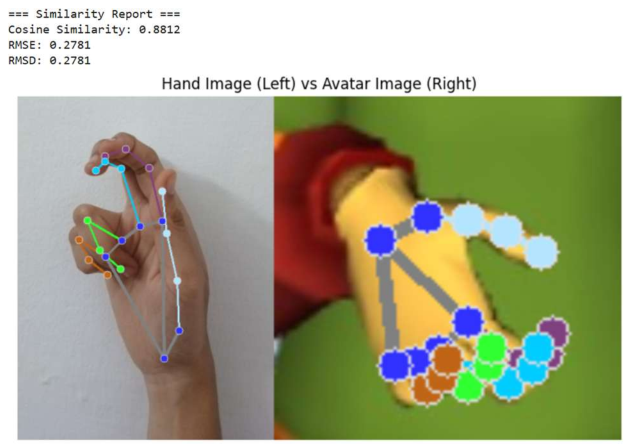


Figure 6: Avatar similarity comparison

4.3.2 Orientation

The palm and finger direction is identified in the palm and finger landmarks of hands and mapped to the HamNoSys orientation symbols by the orientation module.

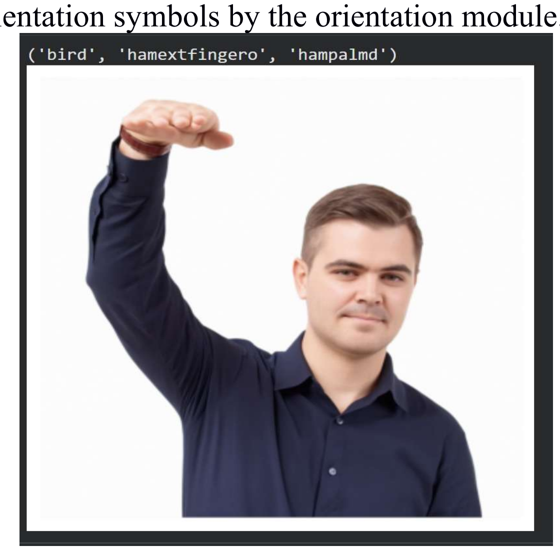


Figure 7: Palm and Finger Orientation Detection Using Hand Landmarks

4.3.3 Location

Head and face Location

The head and face location module works with MediaPipe face detection to process the input frame, which will identify the relative location of the fingertip in the facial area. The system uses normalized coordinates to classify the articulation location and print the HamNoSys label such as hamteeth.

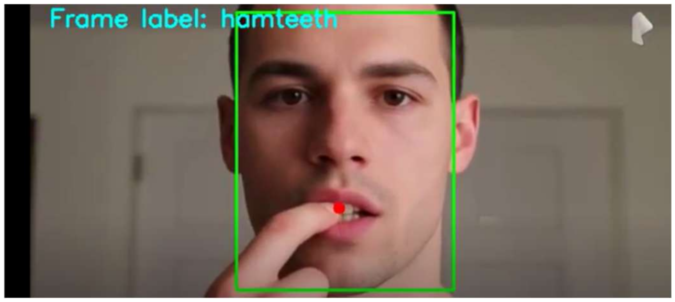


Figure 8: Head and Face Location Classification Using Facial Landmarks

Upper Body Location

The location module of upper body takes the input image and locates the spatial position of the hand in the body in relation to body parts by using body landmarks. The system returns the label of HamNoSys location prediction (e.g., hambelowstomach) as well as a confidence score showing the reliability of the prediction.

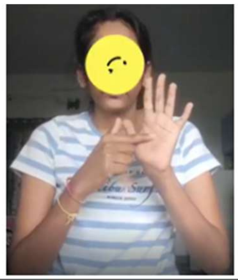
Figure 9: Input frame showing hand position relative to the torso.

```
Predicted: hambelowstomach
Confidence: 0.8
```

Figure 10: Predicted HamNoSys location label *hambelowstomach*.

Hand Location

The hand location module is an analysis that identifies through hand landmarks the input frame and the area of contact that is being touched by the hand. According to landmark positions between frames, the system identifies the dominant hand position and provides the corresponding HamNoSys label like hampalm.



```
Frame counts: Counter({'hampalm': 3, 'hamthumbside': 1})
FINAL HAND LOCATION: hampalm
```

Figure 11: Hand Location Detection Using Hand Landmarks with predicted HamNoSys label *hampalm*.

Finger locations

The finger location module is responsible for identifying labels:
Finger labels: Hamthumb, Hamindexfinger, Hammiddlefinger, Hamringfinger, Hampinky.
Contact based labels- hamfingertip, hamfingermidjoint, hamfingerbase, hambetween
Finger side labels- hamfingernail, hamfingerpad.


```
label = main("/content/drive/MyDrive/pinkyvid.mp4")
print("Final Finger Location Label:", label)

Final Finger Location Label: hampinky
```

Figure 12: Finger Location Identification Using Hand Landmarks with predicted HamNoSys label *hampinky*.

Contact types

The contact types module is responsible for identifying labels - Hamtouch, hamclose, haminterlock, hambrushing, hambehind, hamcross based on the input video.

```
print("Distance:", round(dist(p1, p2), 3))
print("Predicted:", classify_contact(data))

Hands detected: 1
Distance: 0.0
Predicted: hamtouch
```

Figure 13: Contact Type Detection Using Hand Landmarks with predicted HamNoSys contact label *hamtouch*.

Arm and Space Location

The Arm and Signing Space location module manages to recognize the extended-arm articulation as the output with the label hamarmextended depicts. Here the hand is completely in the neutral signing space with the elbow flexion being minimal with the arm of the signer being fully extended as opposed to the torso. The system identifies such arrangement through the calculation of the normalized distance between the wrist impalement and the shoulder and the width of the body, making it independent of the height of the signer and even the camera height. The hybridity of temporal consistency between the frames illustrates that the pose is depicting a rather stationary point of extended arm positioning and not a momentary motion. The cause and effect of neutral space and extended-arm space confusion are observed to be low in experimental observations indicating that the body-normalized thresholding strategy is an effective way to capture large-scale spatial articulation needed by HamNoSys. This finding confirms that the arm and space location module is reliable in the correct encoding of spatial emphasis and semantics of reach related signs.

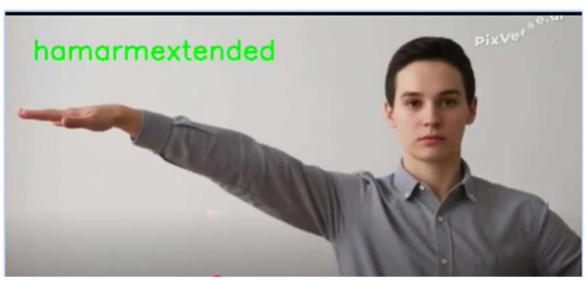


Figure 14: Arm Extension Detection in Signing Space with predicted HamNoSys label *hamarmextended*.

4.3.4 Movement

The awareness of the major direction of the movement of the hand in the process of signing is the work of the Movement module. This consists of such linear directional motions as left, right, up, down, inward, and outward basing on frame-by-frame displacement of hand landmarks. The goal of this module is to encode the overriding instantaneous direction of movement, and not net displacement, and it corresponds with the principles of HamNoSys movement encoding.

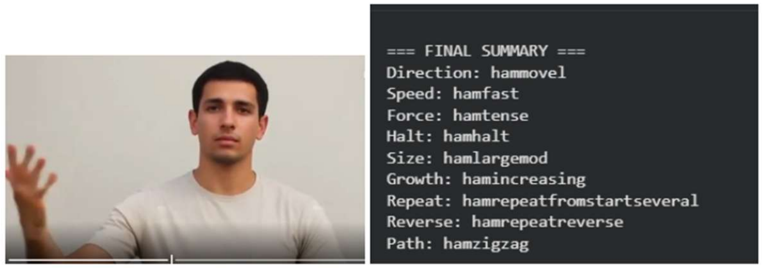


Figure 15: Movement Analysis Module Showing Input Gesture and Predicted HamNoSys Movement Features.

Final Integrated Output

This number reflects the end system output once all modules have been combined and this consists of handshape recognition, orientation detection, location identification, and movement analysis. This system takes the input gesture and produces the HamNoSys labels of each component in terms of landmark analysis and hierarchical prediction. The combined output is a complete symbolic depiction of the sign, which is an indication that all modules have been integrated successfully in the HamNoSys generation pipeline.

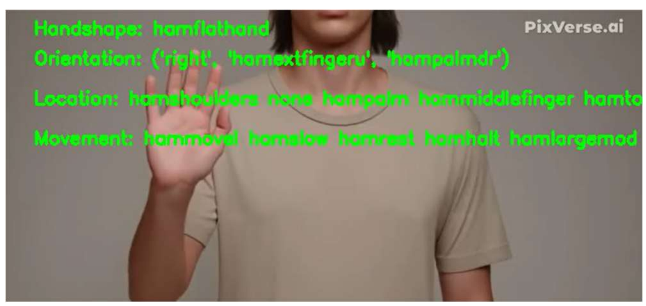


Figure 16: Final Integrated HamNoSys Output Showing Predicted Handshape, Orientation, Location, and Movement Components.

CHAPTER 5
CONCLUSION AND FUTURE ENHANCEMENTS

5.1 SUMMARY OF THE WORK

In this project, a full system was developed of automatically generating HamNoSys notation of sign language images and videos. It is a system that closes the gap between raw gesture input and the sign language representation of symbols by a combination of machine learning and rule-based systems. A personalized set of handshape combinations was generated and augmented by various techniques of augmentation, including rotation, scaling, flipping, and brightness gain to enhance diversity and strength. They were performed with MediaPipe that can extract 21 hand landmarks per image, resulting in 63 numerical features that describe the geometric structure of the hand. Such characteristics were used to train the handshape recognition models.

Handshape recognition in the system is based on a hierarchical classification strategy with various models forecasting base handshape, thumb position, bending, and interaction element of fingers. The reliable predictions made with the help of Random Forest and XGBoost classifiers were the result of selecting the balance between training various categories of hands shapes. The system breaks down the problem into smaller classification tasks as opposed to one large model and this enhances prediction accuracy, as well as eliminating ambiguity between similar hand configurations. The estimated elements are combined in order to produce valid HamNoSys handshape representations.

Besides recognizing handshapes, the system also identifies orientation, location and movement characteristics of the sign. The palm direction and finger alignment are analyzed to determine the orientation, the location is determined through the use of normalized space coordinates to identify the articulation regions like head, face, upper body, and signing space. Movement detection performs temporal analysis of landmark trajectories in order to identify the direction of motion, repeativity and features of trajectory. The modules are applied based on rule-based mathematical formulations built on MediaPipe landmarks, which guarantee maximum accuracy of spatial and motion features interpretation without any need of large annotated datasets.

To be sure it is correct, a validation layer is used to compare the predicted combinations with a reference CSV of the valid HamNoSys symbols and their hexacademic codes. Illegal outputs are reduced and modified to preserve symbolic consistency. Lastly, a verification system based on an avatar translates HamNoSys notation to SiGML and displays the gesture with a signing avatar. This gives the generated notation a symbolic and a visual validation. The general system proves that it is possible to automatically generate HamNoSys with input of signs and this can be used in the future to develop continuous sign language recognition, teaching aids and real-time sign language translation systems.

5.2 CONTRIBUTION IN THE WORK

This work leads to the creation of an automated generation system of HamNoSys notation on sign language input. A new HamNoSys dataset was prepared since there are few publicly available datasets with direct HamNoSys annotations. The data set contains various handshape combinations according to HamNoSys rules and was also diversified by data augmentation methods in order to enhance variety and trainability.

The other valuable contribution is the use of machine learning to recognize hand shapes and orientation, location, and movement detection by rule-based approaches. Such modules apply mathematical formulations that are based on MediaPipe landmarks and anatomically-validated geometric thresholds to effectively interpret gestures. Such a mixture of ML with rule-based methods contributes to the generation of credible HamNoSys representations of sign language gestures.

5.3 FUTURE ENHANCEMENTS

Future research may aim at increasing the dataset by obtaining additional samples of various signers whose hand position, rotation, orientation, and light conditions vary to enhance model strength and precision. The system can further be expanded to facilitate more than one sign language and automatic sign language to text or speech translation, and hence it is more applicable in the real world communications. Also, this would be enhanced by the integration of continuous sign recognition as opposed to isolated signs, which would enable the system to manipulate entire sign sequences. This can be followed up by further development of real-time gesture recognition, better visualization of the avatars to facilitate better validation and implementation through the web or mobile platforms, which would make the system more accessible and useful.

CHAPTER 6

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